

Now, by definition of $\varepsilon_{g,t}$ again,

$$\sum_{t=1}^T N_{g,t} \varepsilon_{g,t} = 0 \text{ for all } g \in \{1, \dots, G\}, \quad (5)$$

$$\sum_{g=1}^G N_{g,t} \varepsilon_{g,t} = 0 \text{ for all } t \in \{1, \dots, T\}. \quad (6)$$

Then,

$$\begin{aligned} & \sum_{g,t} N_{g,t} \varepsilon_{g,t} E(Y_{g,t} | \mathbf{D}) \\ &= \sum_{g,t} N_{g,t} \varepsilon_{g,t} (E(Y_{g,t} | \mathbf{D}) - E(Y_{g,1} | \mathbf{D}) - E(Y_{1,t} | \mathbf{D}) + E(Y_{1,1} | \mathbf{D})) \quad (7) \\ &= \sum_{g,t} N_{g,t} \varepsilon_{g,t} (D_{g,t} E(\Delta_{g,t} | \mathbf{D}) - D_{g,1} E(\Delta_{g,1} | \mathbf{D}) - D_{1,t} E(\Delta_{1,t} | \mathbf{D}) + D_{1,1} E(\Delta_{1,1} | \mathbf{D})) \\ &= \sum_{g,t} N_{g,t} \varepsilon_{g,t} D_{g,t} E(\Delta_{g,t} | \mathbf{D}) \\ &= \sum_{(g,t): D_{g,t}=1} N_{g,t} \varepsilon_{g,t} E(\Delta_{g,t} | \mathbf{D}). \quad (8) \end{aligned}$$

The first and third equalities follow from Equations (5) and (6). The second equality follows from Lemma 1. The fourth equality follows from Assumption 2. Finally, Assumption 2 implies that

$$\sum_{g,t} N_{g,t} \varepsilon_{g,t} D_{g,t} = \sum_{(g,t): D_{g,t}=1} N_{g,t} \varepsilon_{g,t}. \quad (9)$$

Combining (4), (8), (9) yields

$$E(\hat{\beta}_{fe} | \mathbf{D}) = \sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} w_{g,t} E(\Delta_{g,t} | \mathbf{D}). \quad (10)$$

Then, the result follows from the law of iterated expectations.

Proof of Proposition 1

If for all $t \geq 2$ $N_{g,t}/N_{g,t-1}$ does not depend on t , then it follows from the first order conditions attached to Regression 1 and a few lines of algebra that $\varepsilon_{g,t} = D_{g,t} - D_{g,\cdot} - D_{\cdot,t} + D_{\cdot,\cdot}$. Therefore, $w_{g,t}$ is proportional to $D_{g,t} - D_{g,\cdot} - D_{\cdot,t} + D_{\cdot,\cdot}$. Then, for all (g, t, t') such that $D_{g,t} = D_{g,t'} = 1$, $D_{\cdot,t} > D_{\cdot,t'}$ implies $w_{g,t} < w_{g,t'}$. Similarly, for all (g, g', t) such that $D_{g,t} = D_{g',t} = 1$, $D_{g,\cdot} > D_{g',\cdot}$ implies $w_{g,t} < w_{g',t}$.

Proof of Corollary 1

Proof of the first point

We start by proving the first point. If the assumptions of the corollary hold and $\tilde{\Delta}^{TR} = 0$, then

$$\begin{cases} \tilde{\beta}_{fe} &= \sum_{(g,t):D_{g,t}=1} \frac{N_{g,t}}{N_1} w_{g,t} \tilde{\Delta}_{g,t}, \\ 0 &= \sum_{(g,t):D_{g,t}=1} \frac{N_{g,t}}{N_1} \tilde{\Delta}_{g,t}, \end{cases}$$

where the first equality follows from (10). These two conditions and the Cauchy-Schwarz inequality imply

$$|\tilde{\beta}_{fe}| = \left| \sum_{(g,t):D_{g,t}=1} \frac{N_{g,t}}{N_1} (w_{g,t} - 1) (\tilde{\Delta}_{g,t} - \tilde{\Delta}^{TR}) \right| \leq \sigma(\mathbf{w}) \sigma(\tilde{\Delta}).$$

Hence, $\sigma(\tilde{\Delta}) \geq \underline{\sigma}_{fe}$.

Now, we prove that we can rationalize this lower bound. Let us define

$$\tilde{\Delta}_{g,t}^{TR} = \frac{\tilde{\beta}_{fe} (w_{g,t} - 1)}{\sigma^2(\mathbf{w})}.$$

Then,

$$\tilde{\Delta}^{TR} = \sum_{(g,t):D_{g,t}=1} \frac{N_{g,t}}{N_1} \frac{\tilde{\beta}_{fe} (w_{g,t} - 1)}{\sigma^2(\mathbf{w})} = \frac{\tilde{\beta}_{fe}}{\sigma^2(\mathbf{w})} \left(\sum_{(g,t):D_{g,t}=1} \frac{N_{g,t}}{N_1} w_{g,t} - 1 \right) = 0,$$

as it follows from the definition of $w_{g,t}$ that $\sum_{(g,t):D_{g,t}=1} \frac{N_{g,t}}{N_1} w_{g,t} = 1$.

Similarly,

$$\begin{aligned} \sum_{(g,t):D_{g,t}=1} \frac{N_{g,t}}{N_1} w_{g,t} \frac{\tilde{\beta}_{fe} (w_{g,t} - 1)}{\sigma^2(\mathbf{w})} &= \frac{\tilde{\beta}_{fe}}{\sigma^2(\mathbf{w})} \sum_{(g,t):D_{g,t}=1} \frac{N_{g,t}}{N_1} w_{g,t} (w_{g,t} - 1) \\ &= \frac{\tilde{\beta}_{fe}}{\sigma^2(\mathbf{w})} \sum_{(g,t):D_{g,t}=1} \frac{N_{g,t}}{N_1} (w_{g,t} - 1)^2 \\ &= \tilde{\beta}_{fe}, \end{aligned}$$

where the second equality follows again from the fact that $\sum_{(g,t):D_{g,t}=1} \frac{N_{g,t}}{N_1} w_{g,t} = 1$.

Proof of the second point

We first suppose that $\tilde{\beta}_{fe} > 0$. We seek to solve:

$$\begin{aligned} \min_{\tilde{\Delta}_{(1)}, \dots, \tilde{\Delta}_{(n)}} \sum_{i=1}^n \frac{N_{(i)}}{N_1} \left(\tilde{\Delta}_{(i)} - \tilde{\Delta}^{TR} \right)^2 \quad \text{s.t.} \quad \tilde{\beta}_{fe} &= \sum_{i=1}^n \frac{N_{(i)}}{N_1} w_{(i)} \tilde{\Delta}_{(i)}, \\ \tilde{\Delta}_{(i)} &\leq 0 \text{ for all } i \in \{1, \dots, n\}. \end{aligned}$$

This is a quadratic programming problem, with a matrix that is symmetric positive but not definite. Hence, by Frank and Wolfe (1956) and the fact that the linear term in the quadratic problem is 0, the solution exists if and only if the set of constraints is not empty. If $w_{(n)} \geq 0$, the set of constraints is empty because $\sum_{i=1}^n \frac{N_{(i)}}{N_1} w_{(i)} \tilde{\Delta}_{(i)} \leq 0 < \tilde{\beta}_{fe}$. On the other hand, if $w_{(n)} < 0$, this set is non-empty since it includes $(0, \dots, 0, \tilde{\beta}_{fe}/(P_{(n)} w_{(n)}))$.

We now derive the corresponding bound. For that purpose, remark that

$$\sum_{i=1}^n \frac{N_{(i)}}{N_1} \left(\tilde{\Delta}_{(i)} - \sum_{i=1}^n \frac{N_{(i)}}{N_1} \tilde{\Delta}_{(i)} \right)^2 = \sum_{i=1}^n \frac{N_{(i)}}{N_1} \tilde{\Delta}_{(i)}^2 - \left(\sum_{i=1}^n \frac{N_{(i)}}{N_1} \tilde{\Delta}_{(i)} \right)^2.$$

The Karush–Kuhn–Tucker necessary conditions for optimality are that for all i :

$$\begin{aligned} \tilde{\Delta}_{(i)} &= \tilde{\Delta}^{TR} + \lambda w_{(i)} - \gamma_{(i)}, \\ \sum_{i=1}^n \frac{N_{(i)}}{N_1} w_{(i)} \tilde{\Delta}_{(i)} &= \tilde{\beta}_{fe}, \\ \gamma_{(i)} &\geq 0, \\ \gamma_{(i)} \tilde{\Delta}_{(i)} &= 0, \end{aligned}$$

where $\tilde{\Delta}^{TR} = \sum_{i=1}^n \frac{N_{(i)}}{N_1} \tilde{\Delta}_{(i)}$, 2λ is the Lagrange multiplier of the constraint $\sum_{i=1}^n \frac{N_{(i)}}{N_1} w_{(i)} \tilde{\Delta}_{(i)} = \tilde{\beta}_{fe}$ and $2\frac{N_{(i)}}{N_1} \gamma_{(i)}$ is the Lagrange multiplier of the constraint $\tilde{\Delta}_{(i)} \leq 0$.

These constraints imply that $\tilde{\Delta}_{(i)} = 0$ if and only if $\tilde{\Delta}^{TR} + \lambda w_{(i)} \geq 0$. Therefore, if $\tilde{\Delta}^{TR} + \lambda w_{(i)} < 0$, $\tilde{\Delta}_{(i)} \neq 0$ so $\gamma_{(i)} = 0$, and $\tilde{\Delta}_{(i)} = \tilde{\Delta}^{TR} + \lambda w_{(i)}$. Therefore,

$$\tilde{\Delta}_{(i)} = \min(\tilde{\Delta}^{TR} + \lambda w_{(i)}, 0). \quad (11)$$

This equation implies that $\tilde{\Delta}_{(i)} \leq \tilde{\Delta}^{TR} + \lambda w_{(i)}$, which in turn implies that $\tilde{\Delta}^{TR} \leq \tilde{\Delta}^{TR} + \lambda$, so $\lambda \geq 0$.

As a result, $\tilde{\Delta}^{TR} + \lambda w_{(i)}$ is decreasing in i , and because $x \mapsto \min(x, 0)$ is increasing, $\tilde{\Delta}_{(i)}$ is also decreasing in i . Then $\tilde{\Delta}_{(n)} < 0$: otherwise one would have $\tilde{\Delta}_{(i)} = 0$ for all i which would imply $\tilde{\beta}_{fe} = 0$, a contradiction. Let $s = \min\{i \in \{1, \dots, n\} : \tilde{\Delta}_{(i)} < 0\}$. Using again (11), we get

$$\tilde{\Delta}^{TR} = \sum_{i \geq s} \frac{N_{(i)}}{N_1} \tilde{\Delta}_{(i)} = P_s \tilde{\Delta}^{TR} + \lambda S_s.$$

Therefore,

$$\tilde{\Delta}^{TR} = \frac{\lambda S_s}{1 - P_s}. \quad (12)$$

Hence, plugging $\tilde{\Delta}$ in (11), we obtain that for all $i \geq s$,

$$\tilde{\Delta}_{(i)} = \lambda \left\{ \frac{S_s}{1 - P_s} + w_{(i)} \right\}.$$

Finally, using again (11), we obtain

$$\tilde{\beta}_{fe} = \sum_{i \geq s} \frac{N_{(i)}}{N_1} w_{(i)} \tilde{\Delta}_{(i)} = \lambda \left\{ \frac{S_s^2}{1 - P_s} + T_s \right\}.$$

Thus,

$$\lambda = \frac{\tilde{\beta}_{fe}}{T_s + S_s^2/(1 - P_s)}.$$

Then, using what precedes,

$$\begin{aligned} \sigma_{fe}^2 &= \sum_{i \geq s} \frac{N_{(i)}}{N_1} (\lambda w_{(i)})^2 + \sum_{i < s} \frac{N_{(i)}}{N_1} (\tilde{\Delta}^{TR})^2 \\ &= \lambda^2 T_s + (1 - P_s) \left(\frac{\lambda S_s}{1 - P_s} \right)^2 \\ &= \lambda^2 \left[T_s + \frac{S_s^2}{1 - P_s} \right] \\ &= \frac{\tilde{\beta}_{fe}^2}{T_s + S_s^2/(1 - P_s)}. \end{aligned}$$

The result follows, once noted that Equations (11) and (12) imply that $s = \min\{i \in \{1, \dots, n\} : w_{(i)} < -S_{(i)}/(1 - P_{(i)})\}$.

Finally, consider the case $\tilde{\beta}_{fe} < 0$. By letting $\tilde{\Delta}'_{(i)} = -\tilde{\Delta}_{(i)}$ and $\tilde{\beta}'_{fe} = -\tilde{\beta}_{fe}$, we have

$$\sigma_{fe} = \min_{\tilde{\Delta}'_{(1)} \leq 0, \dots, \tilde{\Delta}'_{(n)} \leq 0} \sum_{i=1}^n \frac{N_{(i)}}{N_1} \tilde{\Delta}'_{(i)}^2 - \left(\sum_{i=1}^n \frac{N_{(i)}}{N_1} \tilde{\Delta}'_{(i)} \right)^2 \quad \text{s.t.} \quad \sum_{i=1}^n \frac{N_{(i)}}{N_1} w_{(i)} \tilde{\Delta}'_{(i)} = \tilde{\beta}'_{fe}.$$

This is the same program as before, with $\tilde{\beta}'_{fe}$ instead of $\tilde{\beta}_{fe}$. Therefore, by the same reasoning as before, we obtain

$$\sigma_{fe}^2 = \frac{(\tilde{\beta}'_{fe})^2}{T_s + S_s^2/(1 - P_s)} = \frac{\tilde{\beta}_{fe}^2}{T_s + S_s^2/(1 - P_s)}.$$

Proof of Corollary 2

We have

$$\begin{aligned} \beta_{fe} &= E \left(\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} w_{g,t} \tilde{\Delta}_{g,t} \right) \\ &= E \left(\left(\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} w_{g,t} \right) \tilde{\Delta}^{TR} \right) \\ &= E \left(\tilde{\Delta}^{TR} \right) \\ &= \delta^{TR}. \end{aligned}$$